Quantum Teleportation and Super Dense Coding in Open Quantum Systems

R Afzali and A. H Asghari
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Quantum communication protocols rely critically on the preservation of entanglement and quantum coherence in realistic noisy environments. In this work, we investigate quantum teleportation and dense coding using correlated electron pairs described within a Hubbard–Bogoliubov–de Gennes framework and subjected to non-Markovian open-system dynamics. The superconducting correlations are obtained self-consistently, and the corresponding normal and anomalous Green's functions are employed to construct spatially dependent two-qubit density matrices. The resulting states are evolved using Kraus-operator descriptions of environmental noise, including non-Markovian amplitude damping and dephasing channels. The robustness of the quantum resources is characterized through concurrence and singlet-referenced Uhlmann fidelity, while the Holevo quantity of an equal-probability Pauli encoding ensemble is used to quantify the information available to the corresponding dense-coding protocol. For two representative non-Markovian amplitude-damping regimes, the results reveal a pronounced dependence of quantum correlations on both the spatial separation of the electron pair and the evolution time. Environmental coupling suppresses the initially available entanglement and communication resources, while the nonmonotonic survival probability produces transient revivals of concurrence and singlet fraction. The Pauli-ensemble Holevo quantity evolves differently and approaches the one-bit limit of the long-time amplitude-damped product state. A converged interaction sweep at $U/t = 40$, 75, and 115 shows that the Hubbard interaction changes the initial spatial profiles and hence the subsequent fidelity and Holevo dynamics, but their ordering with U is separation dependent rather than uniformly enhancing or suppressing. All numerical results use the repository's magnitude-amplitude Green's-function reduction; the distinct signed coherence-factor construction is derived but remains to be implemented comparatively.

I. INTRODUCTION

Entanglement converts a shared bipartite state into an operational resource for quantum communication. In quantum teleportation, a maximally entangled pair together with two classical bits enables the transfer of an unknown qubit state without physically transmitting the carrier of that state [3]. In superdense coding, the same elementary resource allows one qubit to convey up to two bits of classical information [4]. In realistic settings, however, quantum resources are generally mixed and interact with their surrounding environments. Their usefulness must therefore be characterized through state-dependent quantities such as concurrence, fidelity, and encoded Holevo information [5, 6, 8].

Superconductors provide a natural many-body setting in which fermionic pairing generates strong quantum correlations between electrons. Within Bardeen–Cooper–Schrieffer and Bogoliubov–de Gennes descriptions, normal and anomalous Green functions encode the spatial structure of these paired states [1, 2]. Reduced two-particle spin states can consequently be reconstructed from the corresponding correlation functions and treated as potential resources for quantum communication. This microscopic approach differs from assuming an idealized Bell pair from the outset, since the resulting entanglement is determined by the underlying electronic dispersion, superconducting pairing symmetry, interaction strength, and electron filling. The spatial dependence of the Green functions further makes it possible to investigate how the available quantum correlations vary with the separation between the two electrons.

Environmental memory introduces an additional layer of complexity. For a qubit coupled to a structured reservoir with a Lorentzian spectral density, the excited-state survival probability can exhibit oscillatory behavior, leading to memory-induced revivals of quantum correlations that would otherwise decay monotonically in a Markovian description [9, 10]. Combining a spatially structured superconducting source with non-Markovian quantum channels therefore provides a framework for investigating how microscopic pairing correlations survive, decay, and re-emerge in the presence of environmental noise. In particular, it allows one to study how the usefulness of the superconducting two-particle state for quantum communication depends simultaneously on spatial separation and evolution time.

Here, we develop and numerically investigate such a framework. We first solve an extended-Hubbard Bogoliubov–de Gennes model on a square lattice and evaluate the associated equal-time normal and anomalous Green functions. These correlation functions are then used to reconstruct normalized two-spin X states for electron pairs separated by different distances and directions in real space. The resulting two-qubit states are subsequently evolved under open-system quantum channels describing independent or correlated environmental interactions, with particular attention to non-Markovian amplitude damping.

The evolution of the superconducting quantum resource is characterized using several complementary quantities. Concurrence is employed to quantify bipartite entanglement, while purity and von Neumann entropy describe the degree of mixedness generated by environmental interactions. Uhlmann fidelity to the initial state monitors the preservation of that state during the

open-system evolution. Separately, the overlap with the spin singlet quantifies the retention of singlet weight; for a pure singlet reference it coincides with the pure-state-referenced Uhlmann fidelity, but it is not automatically an optimized teleportation fidelity. For dense coding, we evaluate the Holevo quantity associated with the standard equal-probability Pauli encoding operations and derive its reduction to an entropy expression involving the two-qubit state and one reduced subsystem.

This framework therefore connects microscopic superconducting correlations with quantities relevant to quantum communication. By examining their joint spatial, angular, temporal, and interaction dependence, we investigate how the structure of the superconducting state and non-Markovian environmental effects influence the persistence and possible revival of entanglement, singlet weight, and dense-coding performance.

II. MODEL AND METHODS

A. Lattice Hamiltonian and self-consistent BdG solution

The superconducting system is described on a two-dimensional square lattice with two helicity- or spin-resolved branches, denoted by $s = \pm 1$. The normal-state dispersion is written as

$$\epsilon_{\mathbf{k}s} = -2t(\cos k_x + \cos k_y) - 2sV_{\mathrm{SO}}\sqrt{\sin^2 k_x + \sin^2 k_y}, \tag{1}$$

where t denotes the nearest-neighbor hopping amplitude and V_{SO} represents the spin-orbit coupling strength.

The superconducting gap contains both on-site and extended-s-wave contributions and is expressed as

$$\Delta_{\mathbf{k}s} = \Delta_0 - \frac{\Delta_s}{4t}\epsilon_{\mathbf{k}s}, \tag{2}$$

where Δ_0 and Δ_s denote the corresponding superconducting order parameters. The momentum dependence of the gap therefore follows directly from the electronic dispersion.

The quasiparticle excitation energy is

$$E_{\mathbf{k}s} = \sqrt{(\epsilon_{\mathbf{k}s} - \mu)^2 + \Delta_{\mathbf{k}s}^2}, \tag{3}$$

where μ is the chemical potential. The corresponding coherence factors are given by

$$u_{\mathbf{k}s}^2 = \frac{1}{2}\left(1 + \frac{\epsilon_{\mathbf{k}s} - \mu}{E_{\mathbf{k}s}}\right), \qquad v_{\mathbf{k}s}^2 = \frac{1}{2}\left(1 - \frac{\epsilon_{\mathbf{k}s} - \mu}{E_{\mathbf{k}s}}\right). \tag{4}$$

The superconducting order parameters and chemical potential are determined self-consistently. The equations solved numerically are

$$0 = \Delta_0 + \frac{1}{8N_k^2}\sum_{\mathbf{k},s} V_{\mathbf{k}s}\frac{\Delta_{\mathbf{k}s}}{E_{\mathbf{k}s}}\tanh\left(\frac{\beta E_{\mathbf{k}s}}{2}\right), \tag{5}$$

$$0 = \Delta_s + \frac{V_1}{N_k^2}\sum_{\mathbf{k},s} \frac{\Delta_{\mathbf{k}s}}{E_{\mathbf{k}s}}\tanh\left(\frac{\beta E_{\mathbf{k}s}}{2}\right), \tag{6}$$

and

$$0 = \frac{1}{2N_k^2}\sum_{\mathbf{k},s} \frac{\epsilon_{\mathbf{k}s} - \mu}{E_{\mathbf{k}s}}\tanh\left(\frac{\beta E_{\mathbf{k}s}}{2}\right) - (1 - n), \tag{7}$$

where n denotes the electronic filling and $\beta = 1/(k_{\mathrm{B}}T)$. The momentum-dependent interaction entering Eq. (5) is

$$V_{\mathbf{k}s} = U + 8V_1 s_{\mathbf{k}} + 4\frac{V_1}{t}V_{\mathrm{SO}} s\lambda_{\mathbf{k}}, \tag{8}$$

with

$$\lambda_{\mathbf{k}} = \sqrt{\sin^2 k_x + \sin^2 k_y}, \qquad s_{\mathbf{k}} = \frac{\cos k_x + \cos k_y}{2}. \tag{9}$$

Here, U denotes the on-site Hubbard interaction strength and V_1 characterizes the extended pairing contribution. In the numerical implementation, V_1 corresponds to the parameter denoted by `Delta_t`. Equations (5)–(7) are solved simultaneously using a multidimensional root-finding procedure. The calculations employ a 100×100 momentum-space grid and are performed at $T = 0$. Unless stated otherwise, the fixed model parameters are $\Delta_t/t = 4.5$, $V_{\mathrm{SO}} = 0$, and $n = 1.875$; thus the two helicity branches coincide in the archived calculations. Energies are expressed in units of t, separations in units of the lattice constant, and times in units of $\hbar/t$.

To examine how the interaction strength modifies the superconducting state, we evaluate the momentum-dependent gap $\Delta_{\mathbf{k}s}$ throughout the Brillouin zone for three representative interaction strengths, $U/t = 40$, 75, and 115. For each interaction strength, Δ_0, Δ_s, and μ are obtained independently from the self-consistent equations before evaluating Eq. (2).

Figure 1 shows that $\Delta_{\mathbf{k}s}$ exhibits a pronounced momentum dependence across the Brillouin zone. This structure follows from the dependence of the gap on $\epsilon_{\mathbf{k}s}$ through Eq. (2). The resulting pattern reflects the extended-s-wave contribution and preserves the symmetry of the underlying square-lattice dispersion.

For $U/t = 40$, the reproduced self-consistent solution gives

$$\frac{\Delta_0}{t} = -0.0207713, \qquad \frac{\Delta_s}{t} = -1.2797794, \qquad \frac{\mu}{t} = 3.4477728, \tag{10}$$

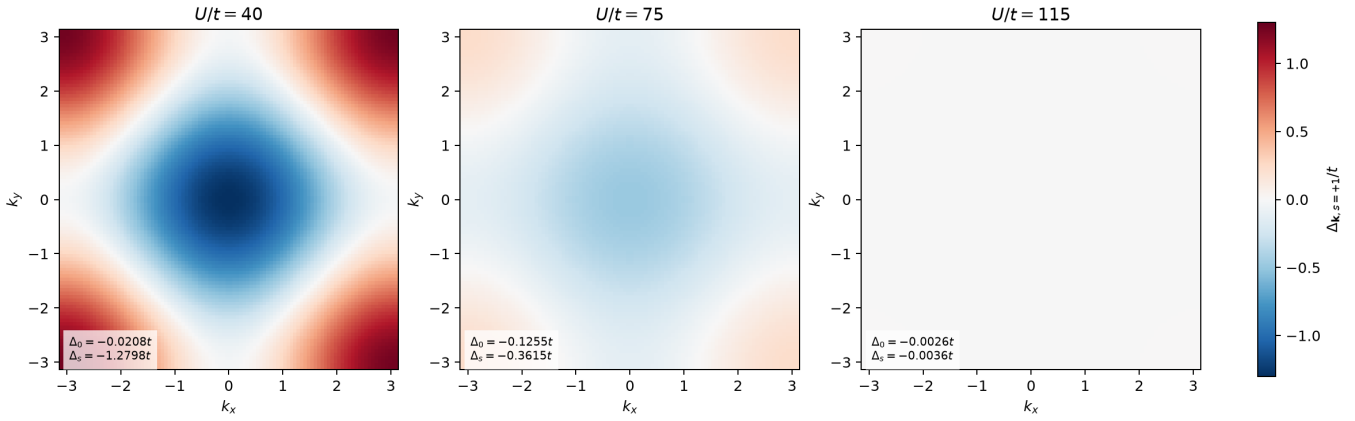


FIG. 1. Momentum-space distribution of the implemented superconducting pairing gap $\Delta_{\mathbf{k},s=+1}$ over the Brillouin zone for $U/t = 40$, 75, and 115, with a common color scale. For each interaction strength, the superconducting order parameters Δ_0 and Δ_s, together with the chemical potential μ, are obtained independently from Eqs. (5)–(7). The fixed parameters are $\Delta_t/t = 4.5$, $V_{\mathrm{SO}} = 0$, $n = 1.875$, $N_k = 100$, and $T = 0$. The momentum dependence originates from the extended-s-wave contribution in Eq. (2).

with maximum absolute residual 2.18×10^{-11}. The relatively large magnitude of Δ_s produces a strong variation of $\Delta_{\mathbf{k}s}$ throughout momentum space. The gap changes sign between different regions of the Brillouin zone because the two terms in Eq. (2) have different momentum dependence.

At the intermediate interaction strength $U/t = 75$, the parameters become

$$\frac{\Delta_0}{t} = -0.1254786, \qquad \frac{\Delta_s}{t} = -0.3614892, \qquad \frac{\mu}{t} = 3.2926234, \tag{11}$$

with maximum absolute residual 2.36×10^{-12}. The reduction in the magnitude of Δ_s suppresses the momentum-dependent variation of the superconducting gap, although the characteristic structure remains visible.

For $U/t = 115$, the solution is

$$\frac{\Delta_0}{t} = -0.00255853, \qquad \frac{\Delta_s}{t} = -0.00355528, \qquad \frac{\mu}{t} = 3.29365466, \tag{12}$$

with maximum absolute residual 2.39×10^{-10}. Both superconducting components are therefore strongly reduced, resulting in a nearly vanishing $\Delta_{\mathbf{k}s}$ throughout the Brillouin zone on the common scale of Fig. 1.

The comparison demonstrates that, for these converged solutions, the interaction strength strongly influences both the magnitude and momentum-space structure of the self-consistent superconducting gap. This behavior is important for the subsequent quantum-information analysis because $\Delta_{\mathbf{k}s}$ enters the quasiparticle spectrum and the normal and anomalous Green functions used to construct the reduced two-particle spin state. Variations in the microscopic pairing structure are therefore transferred to the spatial quantum correlations examined below.

B. Green's functions and the sign of the coherence factor

The normal and anomalous Green's functions are obtained by summing over the two helicity branches and integrating over the Brillouin zone,

$$G_{\uparrow\uparrow}(\mathbf{r}) = -i \int \frac{d^2k}{(2\pi)^2} \sum_s |u_{ks}|^2 e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{13}$$

and

$$F_{\uparrow\downarrow}(\mathbf{r}) = i \int \frac{d^2k}{(2\pi)^2} \sum_s \left(u_{ks} v_{ks}^*\right)^* e^{-i\mathbf{k}\cdot\mathbf{r}}. \tag{14}$$

The corresponding BdG amplitudes satisfy

$$|u_{ks}|^2 = \frac{1}{2}\left(1 + \frac{\epsilon_{ks} - \mu}{E_{ks}}\right), \qquad |v_{ks}|^2 = \frac{1}{2}\left(1 - \frac{\epsilon_{ks} - \mu}{E_{ks}}\right), \tag{15}$$

while the signed anomalous coherence factor obeys the exact BdG identity

$$u_{ks} v_{ks}^* = -\frac{\Delta_{ks}^*}{2E_{ks}}. \tag{16}$$

Two distinct implementations of this coherence factor must be separated. Equation (16) retains the momentum-dependent sign of Δ_{ks} before Brillouin-zone integration. The archived `greens_functions.py`, however, constructs u_{ks} and v_{ks} as non-negative square roots of Eq. (15) and evaluates their product directly. Its numerical anomalous correlator therefore uses an unsigned-amplitude approximation rather than the signed identity in Eq. (16). All archived figures and the new interaction

sweep reported here use that same implemented convention, so they must not be interpreted as a numerical comparison of the signed and unsigned constructions. Such a comparison remains a separate required calculation.

To examine the directional dependence of the implemented spatial correlations, we parameterize the separation vector as

$$\mathbf{r} = r(\cos\theta, \sin\theta), \tag{17}$$

where r denotes the separation between the two electrons and θ specifies the direction relative to the lattice axes. Figure 2 shows the resulting spatial dependence of the magnitudes of the normal and anomalous Green's functions for several representative directions.

As shown in Fig. 2, both implemented Green's functions display oscillatory behavior with an overall decrease in magnitude as the electron separation increases. The comparison among different values of θ reveals directional dependence inherited from the lattice and the self-consistent solution. The symmetry-related directions $\theta = 0^\circ$ and 90° produce identical profiles to numerical precision, as do $\theta = 30^\circ$ and 60°.

A second point concerns the phase structure of the Green's functions. Because ϵ_{ks} and the coherence amplitudes are even functions of $\mathbf{k}$ on the square lattice, integration over the inversion-symmetric Brillouin zone eliminates the odd contribution proportional to $\sin(\mathbf{k}\cdot\mathbf{r})$. Consequently, the Green's functions may be written as

$$G_{\uparrow\uparrow}(\mathbf{r}) = -iA(\mathbf{r}), \qquad F_{\uparrow\downarrow}(\mathbf{r}) = -iB(\mathbf{r}), \tag{18}$$

where $A(\mathbf{r})$ and $B(\mathbf{r})$ are real functions. For the signed construction, we define the normalized correlation functions as

$$g(\mathbf{r}) = \mathrm{Re}\left[\frac{G_{\uparrow\uparrow}(\mathbf{r})}{G_{\uparrow\uparrow}(\mathbf{0})}\right], \qquad f(\mathbf{r}) = \mathrm{Re}\left[\frac{F_{\uparrow\downarrow}(\mathbf{r})}{G_{\uparrow\uparrow}(\mathbf{0})}\right]. \tag{19}$$

These normalized quantities are real in the continuum inversion-symmetric limit. A finite momentum grid can leave a small residual imaginary component, which is discarded by the real-part reduction.

The current numerical density-matrix builder instead uses $|G(\mathbf{r})|/|G(\mathbf{0})|$ and $|F(\mathbf{r})|/|G(\mathbf{0})|$. This distinction requires a precise qualification. The density matrix in Eq. (21) contains only g^2 and f^2; consequently, taking the absolute value of an already real normalized ratio does not change that matrix in exact arithmetic. The physically consequential loss of sign occurs if the sign of Δ_{ks} is discarded *before* the Brillouin-zone sum, as in the unsigned-amplitude implementation described above. It is therefore incorrect to attribute the difference between the two constructions merely to taking an absolute value after integration.

C. Two-spin density matrix

The spatial correlations encoded in the normal and anomalous Green's functions can be used to construct the reduced spin density matrix of two electrons separated by $\mathbf{r}$ [2]. In the basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}, \tag{20}$$

the resulting two-spin state takes the X-state form

$$\rho(\mathbf{r}) = \frac{1}{\mathcal{N}}\begin{pmatrix} 1-g^2 & 0 & 0 & 0 \\ 0 & 1+f^2 & -(g^2+f^2) & 0 \\ 0 & -(g^2+f^2) & 1+f^2 & 0 \\ 0 & 0 & 0 & 1-g^2 \end{pmatrix}, \tag{21}$$

where, for compactness, $g \equiv g(\mathbf{r})$ and $f \equiv f(\mathbf{r})$ denote the normalized normal and anomalous Green's functions defined in Eq. (19). The normalization factor is

$$\mathcal{N} = 4 - 2g^2 + 2f^2, \tag{22}$$

which ensures $\mathrm{Tr}\,\rho(\mathbf{r}) = 1$.

The X structure of Eq. (21) reflects the fact that, for the present correlation functions, the nonvanishing off-diagonal coherence occurs between the opposite-spin states $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. The quantity $g(\mathbf{r})$ describes the contribution associated with normal single-particle correlations, whereas $f(\mathbf{r})$ contains the contribution arising from superconducting pairing. Consequently, the spatial dependence of the microscopic Green's functions is transferred directly to the populations and coherences of the reduced two-spin state.

Equation (21) can equivalently be written in Werner-state form,

$$\rho(\mathbf{r}) = \frac{1-p}{4}\mathbb{I}_4 + p\,|\Psi^-\rangle\langle\Psi^-|, \tag{23}$$

where

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right) \tag{24}$$

is the spin-singlet Bell state and

$$p(\mathbf{r}) = \frac{f^2+g^2}{2+f^2-g^2}. \tag{25}$$

Thus, the parameter $p(\mathbf{r})$ provides a direct connection between the microscopic superconducting correlations and the singlet content of the two-electron spin state. As the separation between the electrons changes, the oscillatory behavior of $g(\mathbf{r})$ and $f(\mathbf{r})$ modifies $p(\mathbf{r})$ and, therefore, the amount of spin entanglement available in the reduced state. In particular, $g(\mathbf{0}) = 1$ implies $p(\mathbf{0}) = 1$, so the implemented zero-separation state is the pure singlet independently of $f(\mathbf{0})$.

The eigenvalues of the Werner state are

$$\lambda_\mathrm{s} = \frac{1+3p}{4}, \qquad \lambda_\mathrm{t} = \frac{1-p}{4}, \tag{26}$$

where λ_s corresponds to the singlet component and λ_t is threefold degenerate for the triplet subspace. These

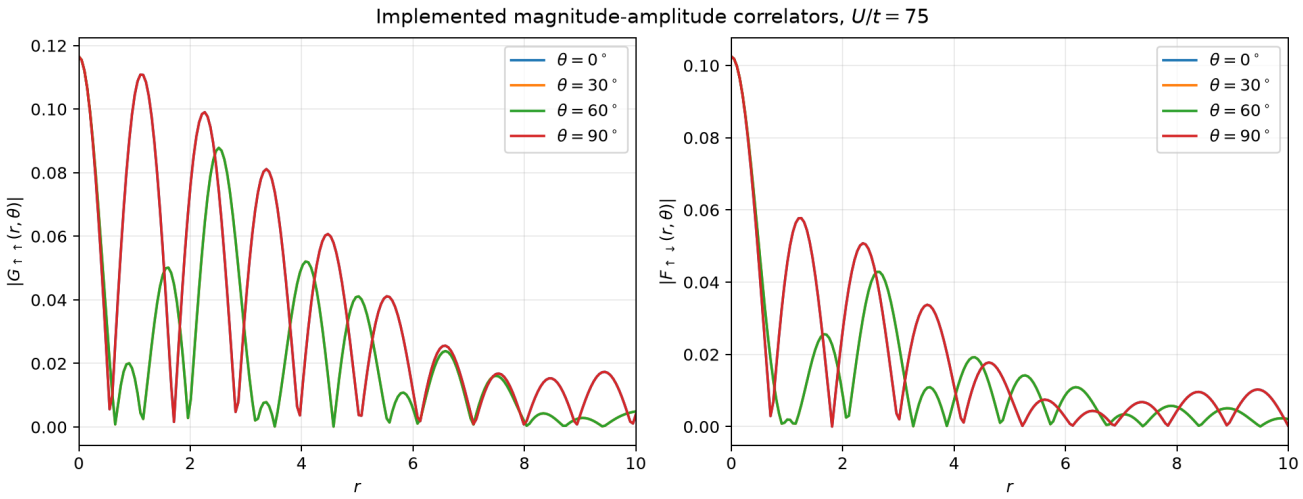


FIG. 2. Angular dependence of the implemented normal and anomalous Green's-function magnitudes at $U/t = 75$ and the initial time. The left panel shows $|G_{\uparrow\uparrow}(r,\theta)|$, while the right panel shows $|F_{\uparrow\downarrow}(r,\theta)|$, for $\theta = 0°$, $30°$, $60°$, and $90°$. Both correlation functions exhibit damped spatial oscillations with direction-dependent amplitudes. The overlapping curves for $0°$ and $90°$, and for $30°$ and $60°$, reflect the square-lattice symmetry. These curves were regenerated from the same magnitude-amplitude implementation used for the archived data.

eigenvalues provide a convenient check of the positivity of the reduced density matrix.

For a Werner state, the concurrence has the closed form

$$C(\mathbf{r}) = \max\left\{0, \frac{3p(\mathbf{r}) - 1}{2}\right\}, \tag{27}$$

showing that the state is entangled when $p > 1/3$. The numerical routine evaluates the X-state reduction of the Wootters concurrence for two-qubit states [5]. For the initial matrix structure in Eq. (21), this expression reduces identically to Eq. (27); the independent amplitude-damping map preserves the X-state structure required by the numerical formula.

D. Open-system evolution

For independent reservoirs, the two-qubit state evolves through local Kraus maps according to

$$\rho(t) = \sum_{\alpha,\beta} \left(K_\alpha^A \otimes K_\beta^B\right) \rho(0) \left(K_\alpha^A \otimes K_\beta^B\right)^\dagger. \tag{28}$$

Here, K_α^A and K_β^B denote the single-qubit Kraus operators associated with the reservoirs coupled to qubits A and B, respectively. The numerical implementation allows the two reservoirs to have different coupling and memory parameters, although identical parameters are used when the two environments are chosen to be equivalent. All six archived figures analyzed below use independent, equivalent amplitude-damping reservoirs.

For the non-Markovian amplitude-damping channel, the single-qubit Kraus operators are

$$K_0(t) = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{p(t)} \end{pmatrix}, \qquad K_1(t) = \begin{pmatrix} 0 & \sqrt{1-p(t)} \\ 0 & 0 \end{pmatrix}, \tag{29}$$

where $p(t)$ is the excited-state survival probability. The lowercase γ is the system–reservoir coupling strength, whereas the uppercase Γ is the reservoir memory rate; they are not interchangeable. For a Lorentzian reservoir in the oscillatory regime, the survival probability is

$$p(t) = e^{-\Gamma t}\left[\cos\left(\frac{dt}{2}\right) + \frac{\Gamma}{d}\sin\left(\frac{dt}{2}\right)\right]^2, \tag{30}$$

with

$$d = \sqrt{2\gamma\Gamma - \Gamma^2}. \tag{31}$$

The oscillatory regime corresponds to $2\gamma > \Gamma$, for which d is real. Both amplitude-damping regimes presented below satisfy this condition.

For completeness, the exact continuation of Eq. (30) to the overdamped regime, $2\gamma \le \Gamma$, is obtained by defining

$$\tilde{d} = \sqrt{\Gamma^2 - 2\gamma\Gamma}, \tag{32}$$

so that

$$p(t) = e^{-\Gamma t}\left[\cosh\left(\frac{\tilde{d}t}{2}\right) + \frac{\Gamma}{\tilde{d}}\sinh\left(\frac{\tilde{d}t}{2}\right)\right]^2. \tag{33}$$

This continuation is present in the standalone fidelity routine. The main concurrence and Holevo evolution routines instead use a critically damped expression outside the oscillatory regime; this discrepancy does not affect

any underdamped result reported here, but it must be resolved before an overdamped study is undertaken.

An independent colored-dephasing channel is also implemented. Its coherence-retention factor is

$$q(t) = \exp\left[-\frac{\gamma}{2}\left(t + \frac{e^{-\Gamma t} - 1}{\Gamma}\right)\right], \tag{34}$$

with the corresponding Kraus operators

$$K_0(t) = \begin{pmatrix} 1 & 0 \\ 0 & q(t) \end{pmatrix}, \qquad K_1(t) = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{1 - q^2(t)} \end{pmatrix}. \tag{35}$$

These operators satisfy the completeness relation

$$\sum_\alpha K_\alpha^\dagger K_\alpha = \mathbb{I}, \tag{36}$$

ensuring a trace-preserving quantum channel.

A collective-dephasing map is also available in the numerical implementation. In the present work, however, all reported results are obtained using independent reservoirs, so that the environmental action on the two qubits is described by the local map in Eq. (28).

E. Fidelity, teleportation, and Pauli-ensemble Holevo information

The fidelity module uses the squared Uhlmann convention

$$\mathcal{F}_{\mathrm{U}}(\rho, \sigma) = \left[\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right]^2. \tag{37}$$

Two references are calculated separately. The initial-state fidelity is

$$\mathcal{F}_{\mathrm{i}}(\mathbf{r}, t) = \mathcal{F}_{\mathrm{U}}[\rho(\mathbf{r}, t), \rho(\mathbf{r}, 0)], \tag{38}$$

which satisfies $\mathcal{F}_{\mathrm{i}}(\mathbf{r}, 0) = 1$ by construction. The singlet-referenced fidelity is

$$\mathcal{F}_{\Psi^-}(\mathbf{r}, t) = \mathcal{F}_{\mathrm{U}}\left[\rho(\mathbf{r}, t), |\Psi^-\rangle\langle\Psi^-|\right]. \tag{39}$$

Because the reference is pure, Eq. (37) reduces exactly to

$$\mathcal{F}_{\Psi^-}(\mathbf{r}, t) = \langle\Psi^-|\rho(\mathbf{r}, t)|\Psi^-\rangle \equiv F_{\mathrm{s}}(\mathbf{r}, t), \tag{40}$$

where F_{s} is the singlet fraction implemented in the observable module. For the initial Werner state, $F_{\mathrm{s}}(\mathbf{r}, 0) = (1 + 3p(\mathbf{r}))/4$.

Neither $\mathcal{F}_{\mathrm{i}}$ nor F_{s} should be identified without further analysis with operational teleportation performance. The latter depends on the fully entangled fraction

$$\mathfrak{f}(\rho) = \max_{\mathcal{B}} \langle\mathcal{B}|\rho|\mathcal{B}\rangle, \tag{41}$$

where the maximum is over the Bell basis after the local optimization appropriate to the teleportation protocol. For a qubit resource, the optimized average teleportation fidelity obeys $\overline{\mathcal{F}}_{\mathrm{tel}} = (2\mathfrak{f}+1)/3$ [7]. For the initial Werner state, $\mathfrak{f} = F_{\mathrm{s}}$ because the singlet eigenvalue in Eq. (26) is the largest Bell weight. No corresponding local optimization is performed for the evolved states in the repository. We therefore do not assign teleportation thresholds to the singlet-fidelity plots below, and the horizontal lines drawn by the plotting routine are visual guides only.

For dense coding, the implemented encoding ensemble consists of four equiprobable messages prepared by

$$\rho_i = (\mathcal{U}_i \otimes I_B)\rho_{AB}(\mathcal{U}_i \otimes I_B)^\dagger, \qquad \mathcal{U}_i \in \{I, \sigma_x, \sigma_y, \sigma_z\}, \tag{42}$$

with $p_i = 1/4$. Here $\mathcal{U}_i$ denotes an encoding unitary, not the Hubbard interaction U. The averaged encoded state is $\bar{\rho}_{AB} = \sum_i \rho_i/4$, and the code calculates

$$\chi_{\mathrm{P}} = S_2(\bar{\rho}_{AB}) - S_2(\rho_{AB}), \qquad S_2(\rho) = -\,\mathrm{Tr}(\rho \log_2 \rho). \tag{43}$$

The subscript P emphasizes the specified Pauli ensemble. The unitaries act on subsystem A after the open-system evolution; no additional encoding or transmission noise is applied by this calculation.

The equal-probability Pauli twirl permits an exact simplification. Writing ρ_{AB} in 2×2 blocks with respect to subsystem A, the four encodings remove the off-diagonal A blocks and equalize the two diagonal blocks. Hence

$$\bar{\rho}_{AB} = \frac{I_A}{2} \otimes \rho_B, \qquad \rho_B = \mathrm{Tr}_A\, \rho_{AB}, \tag{44}$$

and therefore

$$\chi_{\mathrm{P}} = 1 + S_2(\rho_B) - S_2(\rho_{AB}). \tag{45}$$

Equation (45) was also checked directly against the four-unitary implementation. Strong subadditivity gives $0 \le \chi_{\mathrm{P}} \le 2$ bits for a two-qubit state. This is the Holevo quantity of the specified ensemble and the dense-coding capacity when the protocol is restricted to this encoding with ideal subsequent transmission and the corresponding asymptotic joint decoding [8]. It is not, without those conditions, a general optimization over all encodings or a one-shot accessible-information result. The value $\chi_{\mathrm{P}} = 1$ bit is the one-qubit classical benchmark; values above one bit indicate a quantum advantage for this protocol.

A useful limiting check follows from Eq. (45). Independent amplitude damping drives both qubits to their common ground state as $p(t) \to 0$. The resulting pure product state has $S_2(\rho_{AB}) = S_2(\rho_B) = 0$ and therefore $\chi_{\mathrm{P}} \to 1$, even though its concurrence and singlet fraction both vanish. The three quantities consequently need not exhibit the same dynamics.

F. Numerical grids and validation

The archived $\gamma = 1.5$ data use $\gamma_A = \gamma_B = 1.5$, $\Gamma_A = \Gamma_B = 0.1$, 200 radial points on $0 \le r \le 10$, 261 times on $0 \le t \le 80$, and $\theta = 0°, 15°, \ldots, 90°$. The archived

$\gamma = 4.5$ data use $\gamma_A = \gamma_B = 4.5$, $\Gamma_A = \Gamma_B = 0.01$, the same radial and angular grids, and 161 time points on the same time interval. The primary angular slice is 45° in the first regime and 0° in the second. These regimes are analyzed separately below.

Recomputing the initial Green's functions at $U/t = 75$ with the fixed parameters above reproduces the archived $\gamma = 1.5$ arrays to 1.6×10^{-16} maximum absolute error and the $\gamma = 4.5$ arrays to 1.3×10^{-12}. This numerical agreement establishes the model settings of the archived runs, whose NPZ files do not themselves contain a complete parameter record. The initial states are Hermitian and positive semidefinite with unit trace to the tested tolerance. Reconstructed concurrence and singlet-fraction arrays agree with the archived angular data to 2.3×10^{-15}; the singlet-referenced fidelity agrees to 1.5×10^{-8}. The entropy-based Holevo reconstruction agrees with the archived four-Pauli calculation to 3.7×10^{-11}. Additional direct tests of Eq. (28) gave unit trace and Hermiticity, with no negative eigenvalues at the sampled points. The small differences in fidelity are consistent with the separate eigensystem implementation and its eigenvalue clipping.

III. RESULTS AND DISCUSSION

A. Spatial concurrence and interaction dependence

We first examine the spatial structure of the entanglement contained in the two-electron state before the open-system evolution is applied. As shown in Sec. II C, the reduced two-spin density matrix can be written in Werner form with

$$p(\mathbf{r}) = \frac{f^2(\mathbf{r}) + g^2(\mathbf{r})}{2 + f^2(\mathbf{r}) - g^2(\mathbf{r})}. \tag{46}$$

The corresponding concurrence therefore takes the closed form

$$C(\mathbf{r}) = \max\left\{0, \frac{3p(\mathbf{r}) - 1}{2}\right\}, \tag{47}$$

so that the two-spin state is entangled only when $p(\mathbf{r}) > 1/3$. Since both $g(\mathbf{r})$ and $f(\mathbf{r})$ are determined by the microscopic normal and anomalous Green's functions, Eq. (47) directly connects the spatial entanglement to the underlying superconducting correlations.

To investigate how this entanglement is modified by the interaction strength and lattice direction, Fig. 3 shows $C(r, \theta, 0)$ for $U/t = 20$, 44, 79, and 115 and for the representative directions $\theta = 0°$, 30°, 60°, and 90°. These solutions passed the same residual and positivity checks. A previously listed $U/t = 150$ case is not included because the tested solver outputs did not meet the residual criterion and therefore cannot support a converged numerical claim.

Several features are evident from Fig. 3. At $r = 0$, the concurrence is unity for all interaction strengths, as required by $p(\mathbf{0}) = 1$. As the electrons are separated, C decreases rapidly and reaches zero at a finite distance. The precise location of this first entanglement boundary depends on U, demonstrating that the interaction modifies the real-space range over which the implemented superconducting correlations support two-spin entanglement.

The interaction dependence is not a uniform suppression of the concurrence. For the parameters considered here, the $U/t = 20$ curve retains nonzero concurrence to the largest separation within the first connected interval. Increasing U shifts this first zero toward smaller r. In the first axial revival window around $r \simeq 1.1$–1.3, however, the peak concurrence is larger for $U/t = 79$ and 115 than for $U/t = 20$. Thus the interaction changes both the entanglement boundaries and the spatial oscillation amplitudes, and no single ordering applies at every separation.

A pronounced directional dependence is also visible. For $\theta = 0°$ and 90°, the concurrence reappears after the first zero and forms additional entangled windows, including the first strong revival and a second region around $r \simeq 2.1$–2.6. These disconnected regions are a direct consequence of the oscillatory real-space Green's functions: when $p(r, \theta)$ falls below 1/3 the state becomes separable, while subsequent oscillations can drive it above the threshold again.

By contrast, the off-axis revival structure along $\theta = 30°$ and 60° is weaker and displaced to larger r in the displayed interval. The exact agreement between the 0° and 90° results, and between the 30° and 60° results, reflects the square-lattice symmetry. These are spatial revivals of the initial entanglement, not environmental memory effects; the open-system time has not yet been introduced.

B. Non-Markovian concurrence and singlet fidelity

We now consider the regime $\gamma_A = \gamma_B = 4.5$ and $\Gamma_A = \Gamma_B = 0.01$. Figure 4 shows the concurrence along $\theta = 0°$. Entanglement is concentrated in separated spatial islands at the initial time and in repeated temporal windows. Within each temporal window the radial locations remain tied to the initial spatial correlation profile, while the envelope weakens at later times. The repeated windows are consistent with the oscillations of the exact survival probability in Eq. (30); no additional revival mechanism is required to explain them.

The singlet-referenced fidelity in Fig. 5 displays related but nonidentical behavior. It remains nonzero in regions where the concurrence has vanished, as expected because a mixed separable state can retain singlet weight. At $r = 0$, the initial state is a pure singlet and independent amplitude damping gives the direct relations

$$F_{\mathrm{s}}(0, t) = p(t), \qquad C(0, t) = p(t). \tag{48}$$

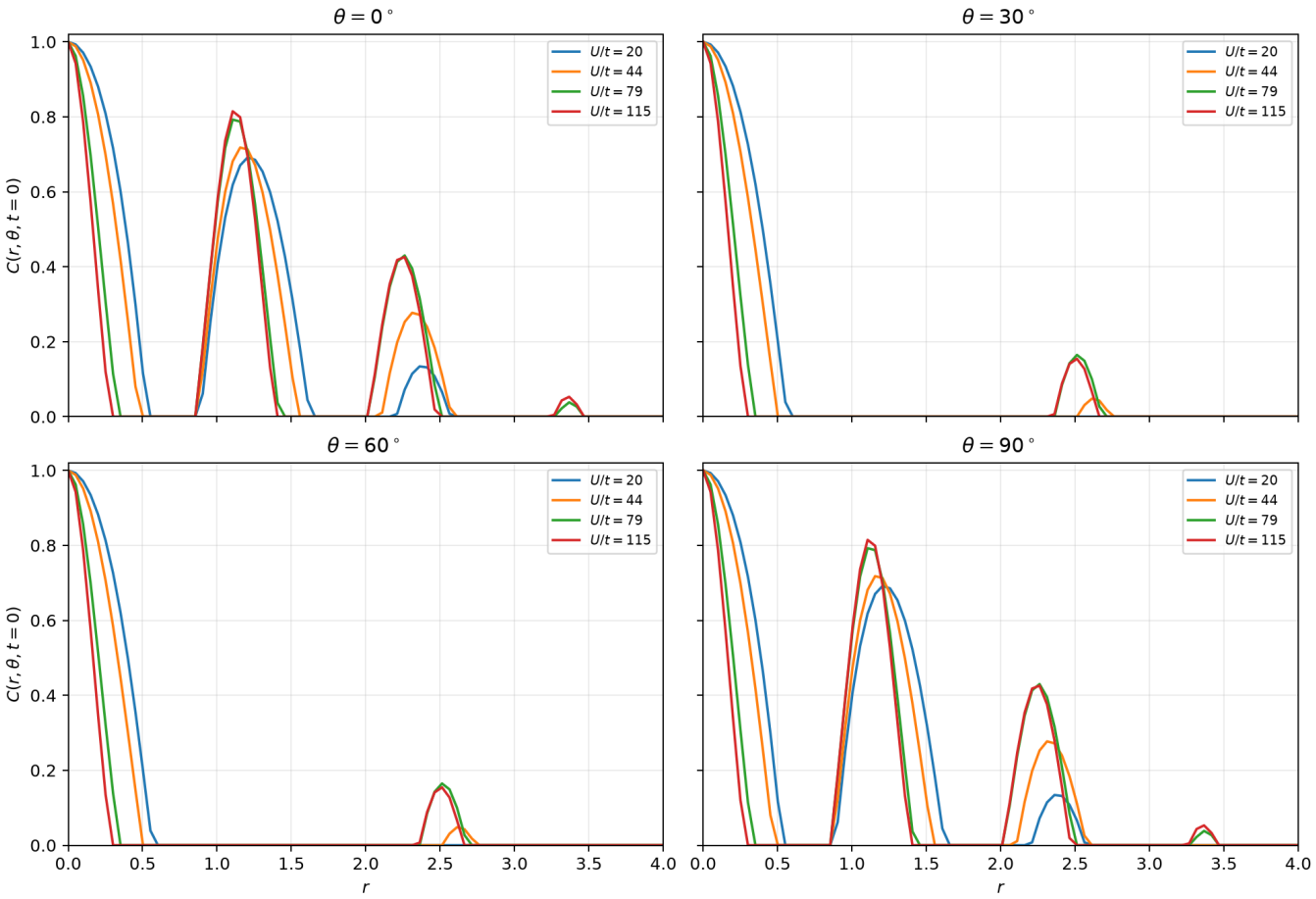


FIG. 3. Initial spatial dependence of the concurrence $C(r, \theta, 0)$ for $U/t = 20$, 44, 79, and 115, evaluated along $\theta = 0°$, 30°, 60°, and 90°. The concurrence equals unity at $r = 0$ and decreases to zero at a U-dependent finite separation. Along the lattice-axis directions, additional finite-r entangled regions appear because of the oscillatory spatial dependence of the implemented Green's functions. Off-axis revival windows are weaker and occur at different separations. Increasing U shortens the first entangled interval in these examples but does not suppress every finite-r revival amplitude.

The two quantities therefore coincide at this special point. At finite separation no such identity holds, and singlet weight can remain nonzero after the X-state concurrence has vanished. The archived time slices give maxima $F_{\mathrm{s}} = 0.81091$, 0.65751, and 0.53309 near $t = 21$, 42, and 63, respectively, followed by a decreasing envelope. At intervening survival-probability minima the singlet fraction nearly vanishes.

The contrast between Figs. 4 and 5 illustrates why concurrence and singlet fraction must be reported separately. Entanglement sudden death occurs whenever the X-state concurrence condition fails, whereas F_{s} can remain finite at finite separation. A nonmonotonic singlet fraction alone is therefore insufficient to establish a revival of entanglement away from the special zero-separation state.

C. Fidelity dynamics at lower coupling

Figure 6 shows the singlet fraction for $\gamma_A = \gamma_B = 1.5$ and $\Gamma_A = \Gamma_B = 0.1$ along $\theta = 45°$. The selected separations are those generated by the plotting routine, $r = 0$, 1.9598, 3.9698, 5.9799, 7.9899, and 10. All curves exhibit damped nonmonotonic behavior associated with the same single-qubit survival probability. At $r = 0$, Eq. (48) remains valid with the new parameters; the first two sampled revival maxima are 0.31136 at $t = 11.6923$ and 0.09694 at $t = 23.3846$. Their decreasing magnitudes exclude any claim of persistent or growing revivals over the computed interval.

The large-r curves begin close to one quarter, consistent with a Werner state whose singlet weight approaches the maximally mixed value $F_{\mathrm{s}} = 1/4$ as $p \to 0$. Their subsequent time dependence is dominated by the common noise channel rather than by a persistent long-range entangled resource. At all displayed separations, the singlet fraction approaches zero at long times because the

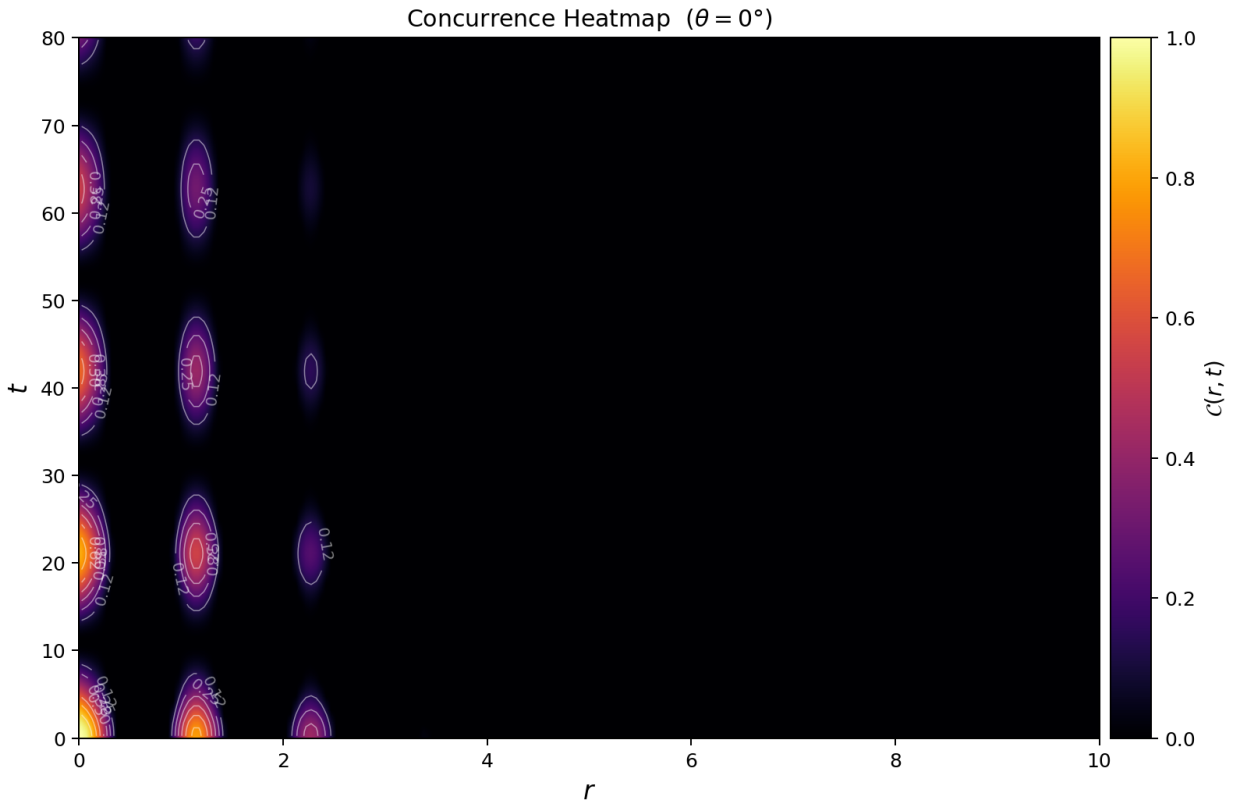


FIG. 4. Concurrence $C(r, 0°, t)$ for independent non-Markovian amplitude damping with $\gamma_A = \gamma_B = 4.5$ and $\Gamma_A = \Gamma_B = 0.01$. The initial state uses $U/t = 75$, $\Delta_t/t = 4.5$, $V_{\rm SO} = 0$, $n = 1.875$, and $N_k = 100$. The axes contain 200 radial points on $0 \le r \le 10$ and 161 time points on $0 \le t \le 80$. Dark regions denote $C = 0$; the separated bright regions are nonzero-concurrence windows generated from the same spatial state and survival probability as the singlet fraction in Fig. 5.

amplitude-damping fixed point is a ground-state product state orthogonal to $|\Psi^-\rangle$. This fixed-point statement does not apply to fidelity to each initial state, which is a different array in the repository.

D. Angular dependence of the singlet fraction

Angular dependence enters the calculation only through $\mathbf{r} = r(\cos\theta, \sin\theta)$ in the Green's functions and hence through $\rho(\mathbf{r}, 0)$. The amplitude-damping Kraus operators and $p(t)$ contain no angle-dependent coefficient. Consequently, the noise does not introduce a new preferred lattice direction; angular structure at finite time is inherited from the initial state and modified by the common temporal map.

Figure 7 displays the initial singlet fraction at seven sampled angles and four separations. The $r = 0$ curve is unity at every angle, as required by $p(\mathbf{0}) = 1$. Nontrivial anisotropy is visible at finite separation. For example, at $r = 3.3166$ the initial singlet fraction ranges from 0.25190 at 30° and 60° to 0.50801 at 45°. At $r = 6.6332$ and 10, the corresponding angular ranges are much smaller, approximately 0.01616 and 0.00154, respectively. The sampled data obey $F_{\rm s}(0°) = F_{\rm s}(90°)$, $F_{\rm s}(15°) = F_{\rm s}(75°)$, and $F_{\rm s}(30°) = F_{\rm s}(60°)$ to numerical precision.

These observations support an initial-state lattice anisotropy at selected finite separations, not a universal preferred direction at all r. Nor does the angle-independent $r = 0$ curve establish isotropy of the finite-r state. Because only the first quadrant was sampled, no independent conclusion about unsampled angles is drawn beyond the lattice symmetries explicitly tested by the available grid.

E. Holevo information and dense coding

The spatial behavior of the Pauli-ensemble Holevo quantity is shown in Fig. 8 for $\theta = 45°$. At $t = 0$, $\chi_{\rm P} = 2$ bits at $r = 0$, as required for a pure singlet. The initial curve then falls below one bit before displaying a finite-separation maximum of 1.06735 bits at $r = 1.6080$. Subsequent sampled maxima at $r = 3.2161$ and 4.8241 are 0.27961 and 0.03566 bits, respectively. Thus the initial spatial profile is nonmonotonic, but only the zero-separation region and the first finite-r maximum exceed the one-bit benchmark in this angular slice.

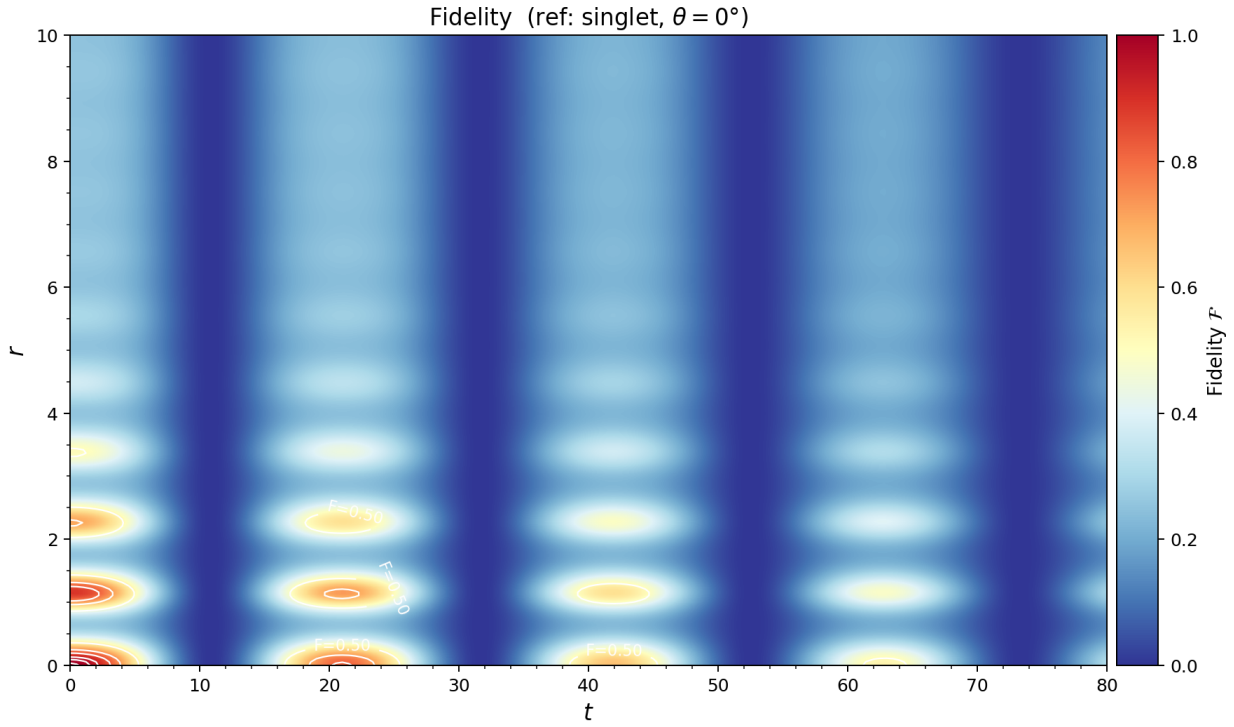


FIG. 5. Singlet-referenced Uhlmann fidelity $\mathcal{F}_{\Psi^-}(r, 0^\circ, t) = F_{\mathrm{s}}(r, 0^\circ, t)$ for the same independent amplitude-damping regime and model settings as Fig. 4. The color scale is the overlap with $|\Psi^-\rangle$, not fidelity to the initial mixed state and not an optimized average teleportation fidelity. The radial oscillations are inherited from the initial two-spin state, while the repeated temporal modulation follows $p(t)$. Contour lines are plotting guides and are not assigned operational teleportation thresholds.

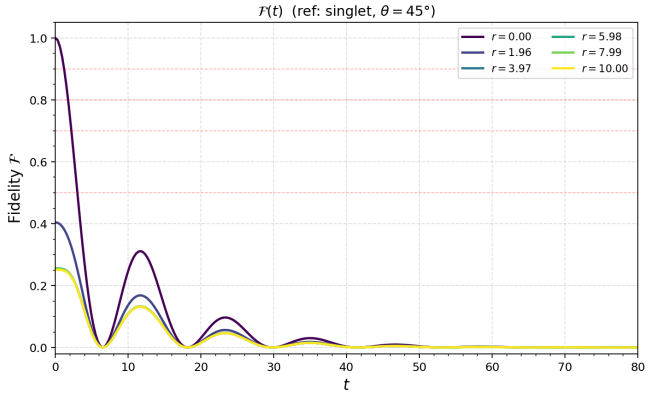


FIG. 6. Time dependence of the singlet-referenced fidelity $\mathcal{F}_{\Psi^-}(r, 45^\circ, t) = F_{\mathrm{s}}$ for independent amplitude damping with $\gamma_A = \gamma_B = 1.5$ and $\Gamma_A = \Gamma_B = 0.1$, using the common $U/t = 75$ initial-state settings. The data contain 261 times on $0 \le t \le 80$. The horizontal dashed lines at 0.5, 0.7, 0.8, and 0.9 are automatically drawn reference guides; they are not thresholds for the plotted singlet fraction.

The temporal slices flatten toward one bit, with small finite-time deviations still visible in the arrays. This tendency follows from the amplitude-damping fixed point derived below Eq. (45), rather than from a survival of dense-coding quantum advantage. Indeed, the asymptotic value one is precisely the classical one-qubit benchmark. At $t = 16$, for example, the sampled radial range is 0.79509–0.86087 bits, below that benchmark throughout this slice.

Figure 9 resolves the angular and temporal behavior at $r = 0$. The initial value is two bits at every sampled angle. Subsequent oscillations are identical across the seven angular rows to numerical precision. This angle independence is expected because the zero-separation initial state is the same pure singlet for every row and the independent noise channel is angle independent. The figure therefore verifies the consistency of the $r = 0$ slice; it does not demonstrate that χ_{P} is angle independent at finite r, where the initial state is anisotropic.

The Holevo dynamics differ markedly from the singlet-fidelity dynamics in Fig. 6. Whereas the singlet fraction tends to zero with the ground-state product fixed point, χ_{P} tends to one bit. During intermediate times it can fall below one and subsequently return toward one as the state becomes less mixed. Such an increase toward the classical benchmark must not be described as a revival of entanglement or as an increase above the dense-coding classical limit. More generally, Eq. (45) shows that the dynamics are controlled by both the global entropy and the subsystem entropy, so identical behavior of concurrence, singlet fraction, and Holevo information is not expected.

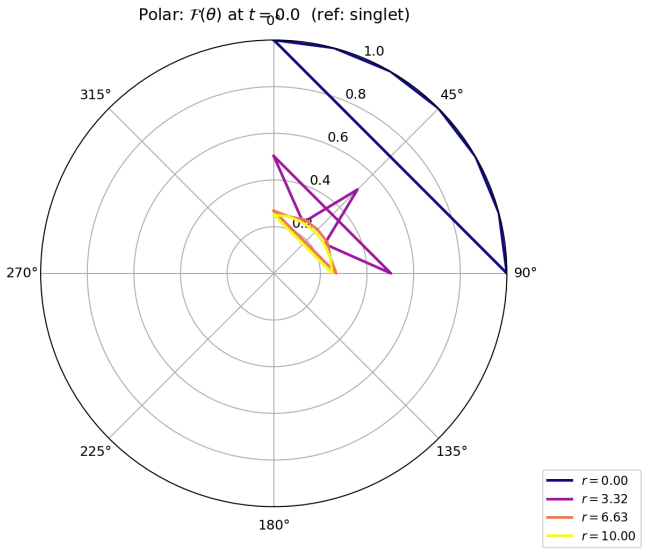


FIG. 7. Polar representation of the initial singlet fraction $F_{\mathrm{s}}(r, \theta, 0)$ for $r = 0$, 3.3166, 6.6332, and 10 in the $\gamma = 1.5$, $\Gamma = 0.1$ data set. Only $\theta = 0°, 15°, \ldots, 90°$ were simulated. The plotted closure between 90° and 0° is a line connecting the sampled endpoints, and the unpopulated remainder of the polar circle must not be interpreted as additional angular data. The radial coordinate is the singlet fraction, not the electron separation.

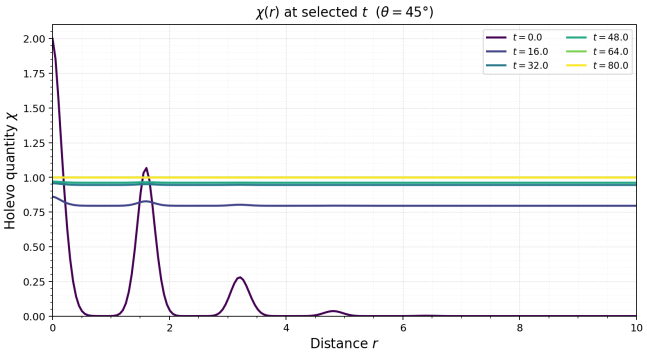


FIG. 8. Pauli-ensemble Holevo quantity $\chi_{\mathrm{P}}(r, 45°, t)$ at $t = 0, 16, 32, 48, 64, 80$ for independent amplitude damping with $\gamma_A = \gamma_B = 1.5$, $\Gamma_A = \Gamma_B = 0.1$, and the common $U/t = 75$ initial-state settings. Entropies use base-two logarithms. The quantity is defined by Eqs. (43) and (45); it is not concurrence or singlet fidelity.

F. Hubbard interaction dependence of fidelity and Holevo information

The Hubbard interaction enters Eq. (8), changes the self-consistent triple $(\Delta_0, \Delta_s, \mu)$, and thereby modifies $E_{\mathbf{k}s}$, the Green's functions, and the initial Werner parameter. The open-system Kraus operators themselves contain no U. Consequently, any computed dependence $\mathcal{F}(U, \theta, r, t)$ or $\chi_{\mathrm{P}}(U, \theta, r, t)$ is transmitted through the initial state and then acted upon by a U-independent noise map.

To test this dependence numerically, we performed a new sweep at the three interaction strengths already used in Fig. 1: $U/t = 40$, 75, and 115. All other model parameters were fixed, and each self-consistency solution satisfied the residual criterion reported in Sec. II F. For each interaction strength we used the full seven-angle, 200-radius, and 261-time grid in the $\gamma_A = \gamma_B = 1.5$, $\Gamma_A = \Gamma_B = 0.1$ independent-amplitude-damping regime. Concurrence, singlet fraction, and the Pauli-ensemble Holevo quantity were evaluated with the same validated formulas. The initial density matrices were Hermitian, normalized, and positive semidefinite. The complete arrays, self-consistency parameters, residuals, and commit identifier are stored in the accompanying NPZ file.

The upper panels of Fig. 10 show that U modifies both the positions and amplitudes of the spatial structures. The ordering is separation dependent. At $r = 0.50251$, for example, the initial singlet fraction decreases from 0.53243 at $U/t = 40$ to 0.350996 at 75 and 0.284845 at 115; the corresponding Holevo values are 0.26196, 0.03640, and 0.00454 bits. At $r = 1.50754$, by contrast, the singlet fraction increases across the same three interactions, taking values 0.717745, 0.778551, and 0.783515, while χ_{P} takes values 0.694131, 0.886208, and 0.903183 bits. These paired examples directly rule out a universal monotonic enhancement or suppression over the spatial domain.

The lower panels show the consequences at fixed $r = 1.50754$. The singlet fraction exhibits damped revivals at common times because $p(t)$ is shared; the interaction changes the initial amplitude and therefore the detailed curve, but does not introduce a separate environmental timescale. The Holevo curves likewise approach one bit at long times, with nearly indistinguishable late-time behavior on the plotted scale. At $r = 0$, all three interactions instead give exactly the same curves because the zero-separation initial state is the pure singlet for every converged solution.

The supported conclusion is therefore specific: within the three-point, fixed-filling, magnitude-amplitude calculation, the Hubbard interaction changes the initial spatial correlations and thereby changes the subsequent singlet-fidelity and Pauli-Holevo dynamics. The data do not establish a global functional law in U, an optimal interaction strength, or a universally beneficial interaction effect. A denser sweep and a comparative implementation of the signed coherence factor would be required before extending these conclusions beyond the present implementation and parameter set.

IV. CONCLUSION

We have examined the Green's-function construction of two-spin BCS density matrices and their subsequent independent non-Markovian amplitude-damping dynamics. The exact signed BdG identity $u_{\mathbf{k}s} v^*_{\mathbf{k}s} = -\Delta^*_{\mathbf{k}s}/2E_{\mathbf{k}s}$

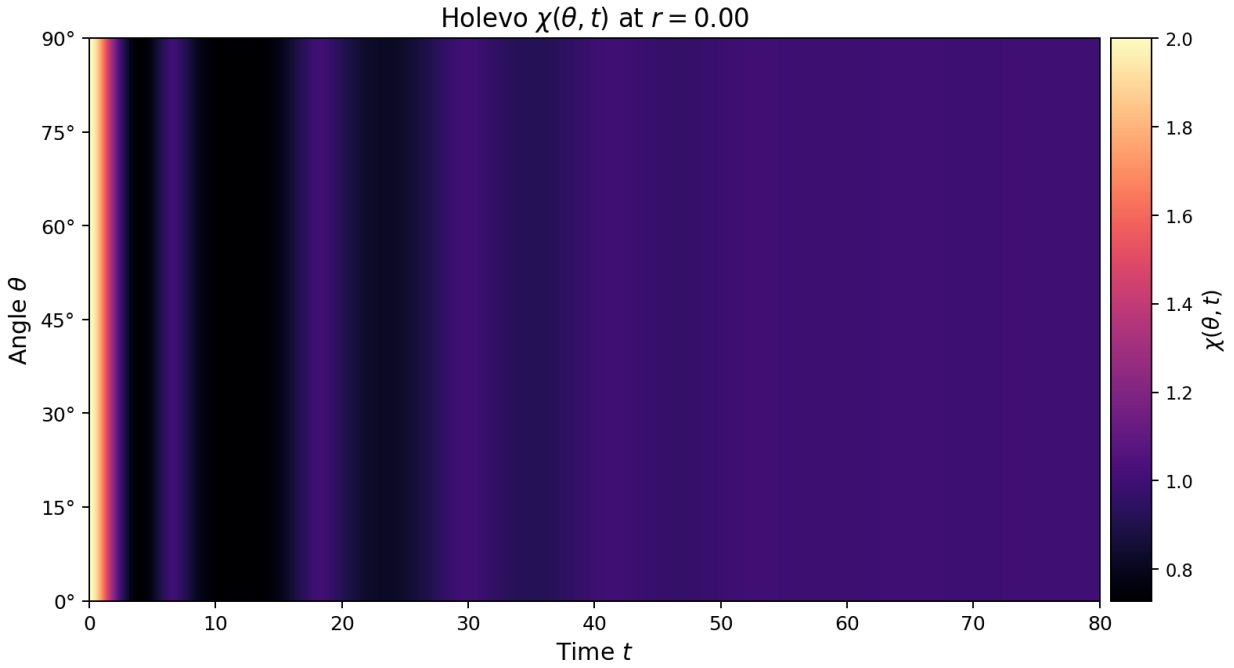


FIG. 9. Pauli-ensemble Holevo quantity $\chi_{\mathrm{P}}(\theta, r = 0, t)$ for the same $\gamma = 1.5$, $\Gamma = 0.1$ regime as Fig. 8. The seven rows correspond to $\theta = 0°, 15°, \ldots, 90°$, and 261 time points span $0 \leq t \leq 80$. The identical rows follow from the angle-independent pure-singlet initial state at $r = 0$ and the angle-independent local Kraus map; no finite-separation isotropy is implied.

must be distinguished from the repository's current unsigned-amplitude numerical implementation. We have also shown that taking the absolute value of an already real normalized Green's-function ratio does not itself alter the present density matrix, which depends only on g^2 and f^2; the physically important difference is whether the gap sign is retained before momentum integration.

The archived results establish spatially structured concurrence and singlet fraction, with temporal revivals determined by the oscillatory survival probability in the two underdamped regimes considered. Singlet-referenced Uhlmann fidelity is exactly the singlet fraction, but it is not an optimized teleportation fidelity for the evolved states. The equal-probability Pauli encoding satisfies $\bar{\rho}_{AB} = (I_A/2) \otimes \rho_B$ and hence $\chi_{\mathrm{P}} = 1 + S_2(\rho_B) - S_2(\rho_{AB})$. Its approach to one bit at long times reflects the pure product fixed point and does not represent a residual dense-coding quantum advantage. Angular structure is inherited from the initial Green's functions, not introduced by the angle-independent noise.

Finally, the new converged sweep at $U/t = 40$, 75, and 115 demonstrates interaction-dependent modifications of the initial and evolved information measures within the implemented convention. Their ordering changes with separation, precluding a universal claim of interaction-induced enhancement or suppression. The principal remaining tasks are a direct signed-versus-unsigned Green's-function comparison, a unified parameter-preserving evolution pipeline for all observables, and locally optimized fully entangled fractions if operational teleportation fidelities are to be assigned to the evolved states.

ACKNOWLEDGMENTS

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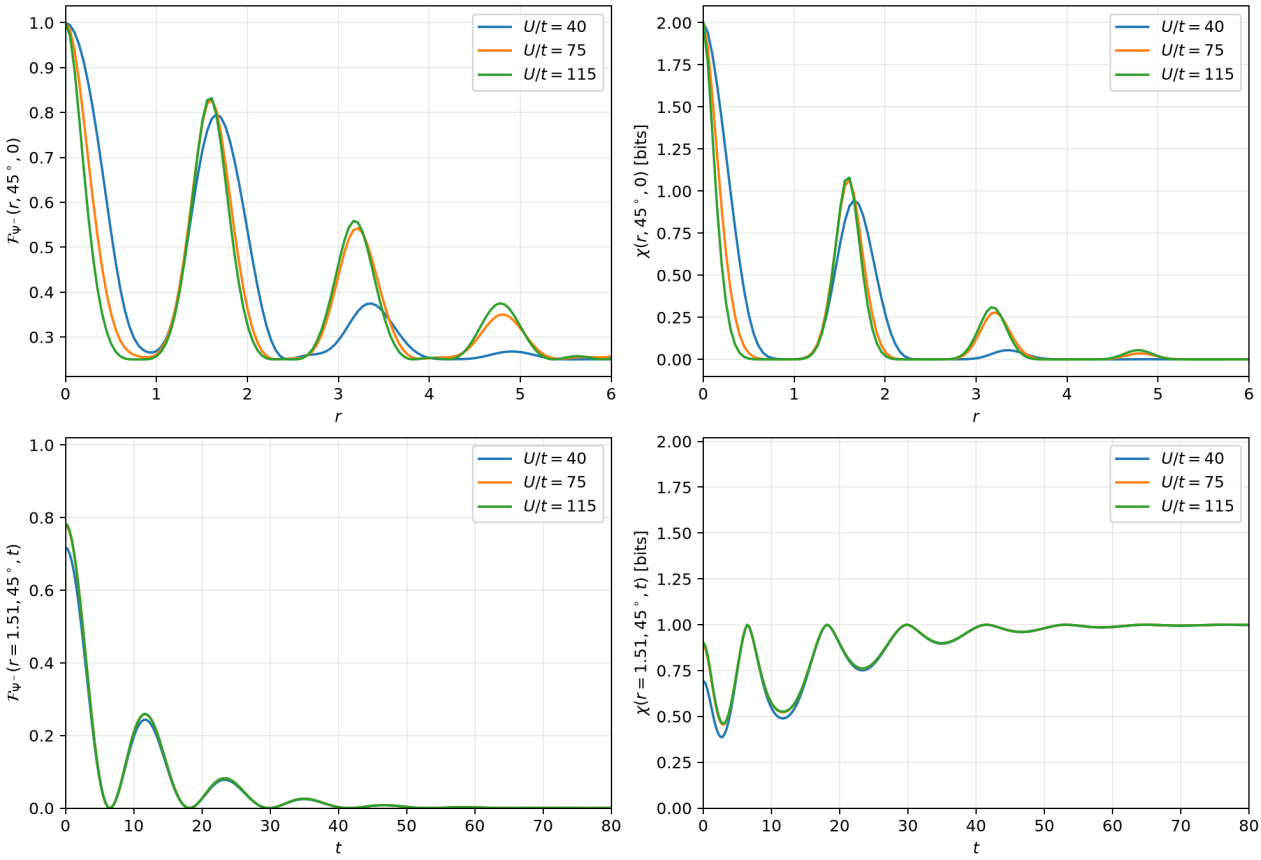


FIG. 10. Interaction dependence computed for $U/t = 40$, 75, and 115 in the $\gamma_A = \gamma_B = 1.5$, $\Gamma_A = \Gamma_B = 0.1$ regime. Upper panels: initial singlet fraction and Pauli-ensemble Holevo quantity along $\theta = 45°$. Lower panels: their time dependence at the fixed sampled separation $r = 1.50754$ and the same angle. All other BdG parameters are identical to those in Fig. 1. The interaction changes the initial state but does not enter the Kraus map. These results use the repository's magnitude-amplitude Green's-function convention.

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